Reasoning Divergence and Cross-Lingual Correctness: A Logistic Regression Analysis

1 Question

Does a larger Reasoning Divergence Score (RDS) between English and Hindi reasoning traces go with a higher chance that the two languages get different correctness outcomes on the same question? All results come from existing Sarvam-105B outputs. No model was retrained or rerun.

2 Method

Outcome. For each English and Hindi question pair we define

$$\text{disagree} = \begin{cases} 0 & \text{both correct or both incorrect,} \\ 1 & \text{exactly one language is correct.} \end{cases}$$

Predictor. RDS is the normalized edit distance between the English and Hindi reasoning traces, ranging from 0 (identical) to 1 (completely different).

Models. We fit logistic regressions with standard errors clustered by question, since each question appears in up to three insertion variants and both reasoning modes.

- **Primary:** disagree $\sim$ RDS + difficulty + task + insertion type + reasoning mode.
- **Length-adjusted:** the primary model plus a trace length gap, $|\ell_{en} - \ell_{hi}| / \max(\ell_{en}, \ell_{hi})$.
- **Per-mode:** the RDS slope estimated separately for reasoning OFF and reasoning ON.

Reference levels are short difficulty, the `date_addition` task, no insertion, and reasoning OFF.

Sample. Of 21,600 loaded pairs, 938 (4.3%) were dropped because the English or Hindi answer could not be parsed. The analysed sample has **20,662 pairs** covering 6,739 unique questions: 19,471 with reasoning OFF and 1,191 with reasoning ON.

3 Results

3.1 How often do the languages disagree?

English and Hindi disagreed on **31.4%** of pairs (95% CI [30.7, 32.0]). The disagreement is lopsided: 92% of cases are English correct and Hindi wrong.

Table 1: Correctness outcomes by reasoning mode.

Mode	Both correct	EN only	HI only	Both wrong	Disagree rate
All	12,468	5,967	516	1,711	31.4%
OFF	11,309	5,947	505	1,710	33.1%
ON	1,159	20	11	1	2.6%

Disagreement rises with difficulty, from 26.7% on short questions to 38.6% on very very long ones. It varies most by task: `date_recurrence` (65.1%) and `day_of_week` (55.2%) are highest, while `date_duration` (12.7%) and `date_addition` (14.9%) are lowest. Insertion type has almost no effect (30.7% to 32.2%).

3.2 Is RDS higher when the languages disagree?

Yes. Mean RDS is 0.809 when the languages agree and 0.917 when they disagree (medians 0.828 and 0.933). A Mann-Whitney test confirms the gap ($p < 0.001$): a randomly chosen disagreeing pair has a higher RDS than a randomly chosen agreeing pair 80% of the time.

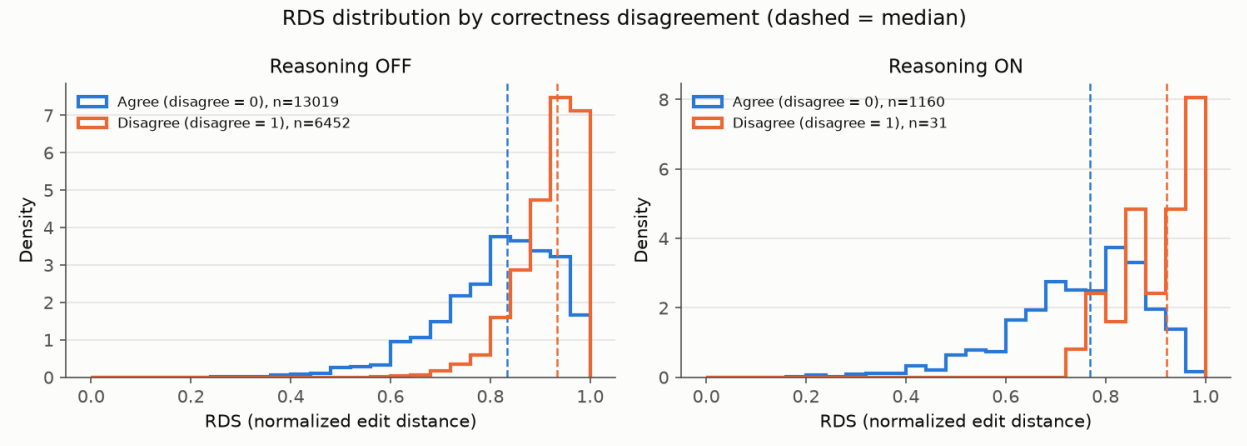


Figure 1: RDS distribution for agreeing and disagreeing pairs. Dashed lines mark medians.

3.3 Regression estimates

Table 2: RDS coefficient across models. OR and AME are per 0.1 increase in RDS. AME is the average change in disagreement probability in percentage points.

Model	Sample	n	β	OR [95% CI]	AME (pp)	Reliable
Primary	All	20,662	15.94	4.93 [4.58, 5.30]	22.9	yes
Length-adjusted	All	20,662	17.20	5.58 [5.18, 6.03]	24.5	yes
OFF, adjusted	OFF	19,471	15.83	4.87 [4.53, 5.24]	24.0	yes
OFF, RDS only	OFF	19,471	13.92	4.02 [3.81, 4.25]	24.0	yes
ON, adjusted	ON	1,191	24.67	11.79 [4.90, 28.37]	4.6	no
ON, RDS only	ON	1,191	20.46	7.74 [3.49, 17.18]	4.6	yes

All RDS coefficients are positive with $p < 0.001$.

Primary model. Holding difficulty, task, insertion type and reasoning mode fixed, a 0.1 increase in RDS multiplies the odds of disagreement by **4.93**. On the probability scale this averages to about 23 percentage points per 0.1 of RDS. The effect is not linear: predicted disagreement stays near zero below RDS 0.7 and climbs steeply above 0.8 (Figure 2).

Length-adjusted model. Edit distance can grow just because one trace is longer. Adding the length gap does not weaken the result: the RDS odds ratio rises slightly to **5.58**. The length gap itself has a negative coefficient ($\beta = -1.19$, $p < 0.001$), so trace length differences do not explain the association.

Reasoning ON vs OFF. Disagreement is far lower with reasoning ON (2.6%) than OFF (33.1%), and reasoning ON lowers the adjusted odds of disagreement by about 90% (OR 0.10). The RDS slope is positive in both modes. However, reasoning ON has only 31 disagreements. Its fully adjusted model did not converge because the `time_duration` task had zero disagreements, so only its RDS-only model should be read, and with caution.

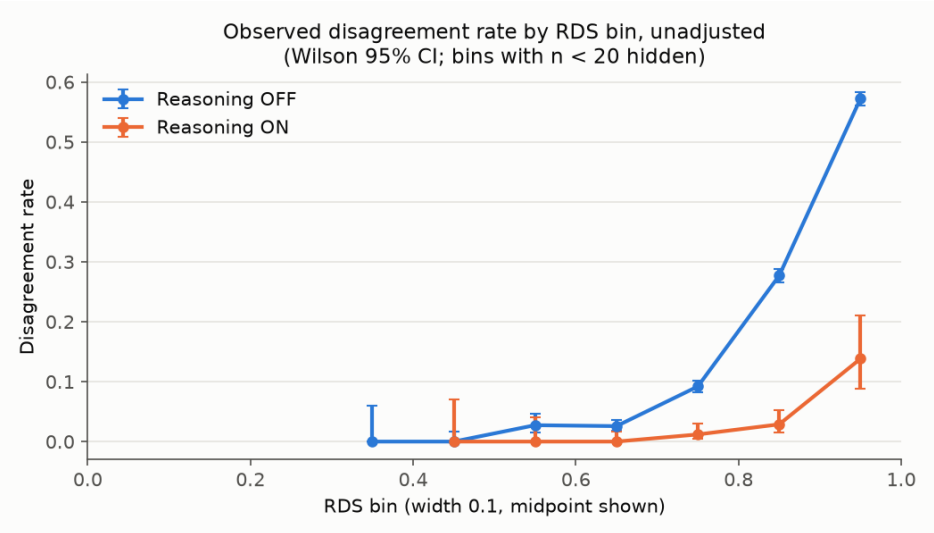


Figure 2: Observed disagreement rate by RDS bin (unadjusted, Wilson 95% CI).

4 Limitations

- **Observational.** RDS and correctness come from the same generations, so we cannot say divergence causes errors. Both may reflect question difficulty not captured by the controls.
- **Mode imbalance.** Reasoning OFF makes up 94% of the pooled sample, so the pooled slope mostly describes reasoning OFF.
- **Asymmetric outcome.** Since 92% of disagreements are Hindi failures, the outcome mainly tracks Hindi-specific errors.
- **Narrow RDS range.** 88% of pairs have RDS above 0.7, so the slope is estimated mainly from the upper range.
- **Measurement.** RDS depends on the step parser and edit-distance costs. Pairs with unparseable answers were dropped.
- **In-sample.** The models describe associations in this dataset, not predictive accuracy on new questions.

5 Conclusion

Greater divergence between English and Hindi reasoning traces is strongly associated with the two languages reaching different correctness outcomes. The association holds after controlling for difficulty, task, insertion type, reasoning mode and trace length. RDS is therefore a useful signal of cross-lingual reasoning inconsistency, though the analysis is correlational and does not show that divergence causes the errors.